MLOPS ASSIGNMENT – 2

NAME : Chandana K T

USN : 4GW23CI009

1. Which run had the lowest latency, and why might that be?

run_batch8 had the lowest latency at 0.120 seconds, followed by run_batch4 at 0.133 seconds and run_batch1 at 0.226 seconds. A larger batch size allows the model to process multiple sentences in a single batch, reducing processing overhead per sentence. The first run may also take longer because of model warm-up overhead.

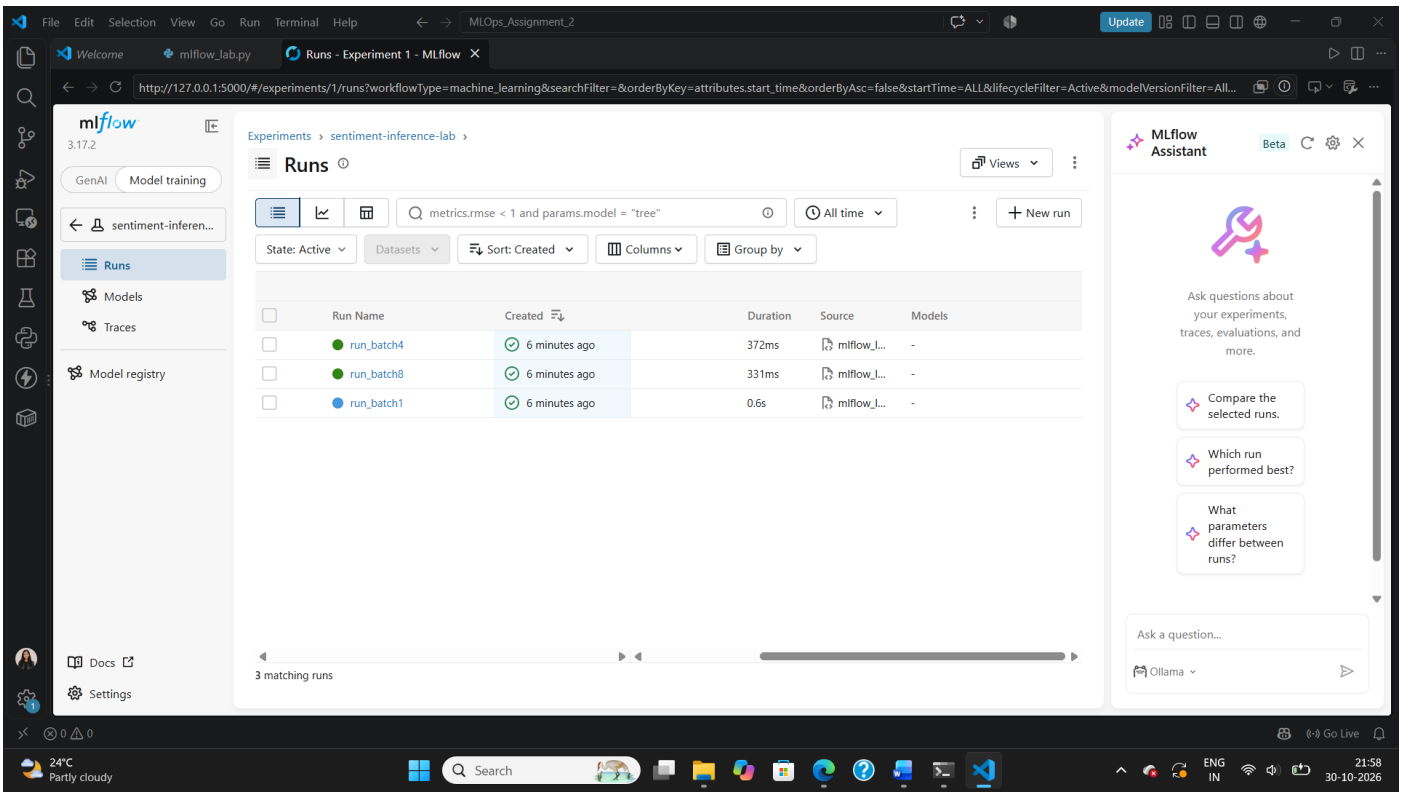
2. Did accuracy change between runs? Why or why not?

No, accuracy remained at 1.0 (100%) across all three runs. Each run used the same pre-trained sentiment analysis model and the same 10 test sentences. Since only the batch size changed, the predicted labels remained the same, so accuracy did not change.

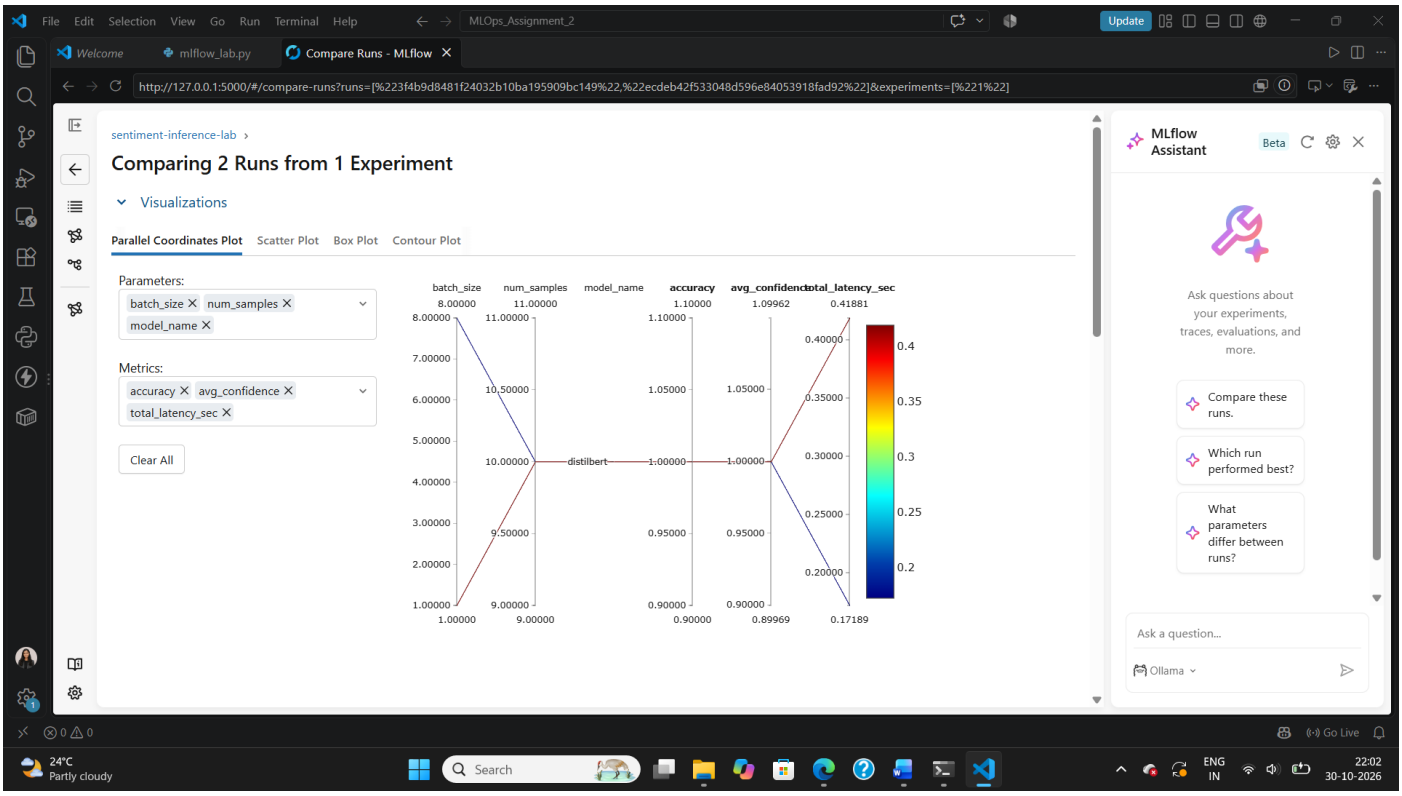
3. Name two benefits of using MLflow here instead of printing results in the terminal.

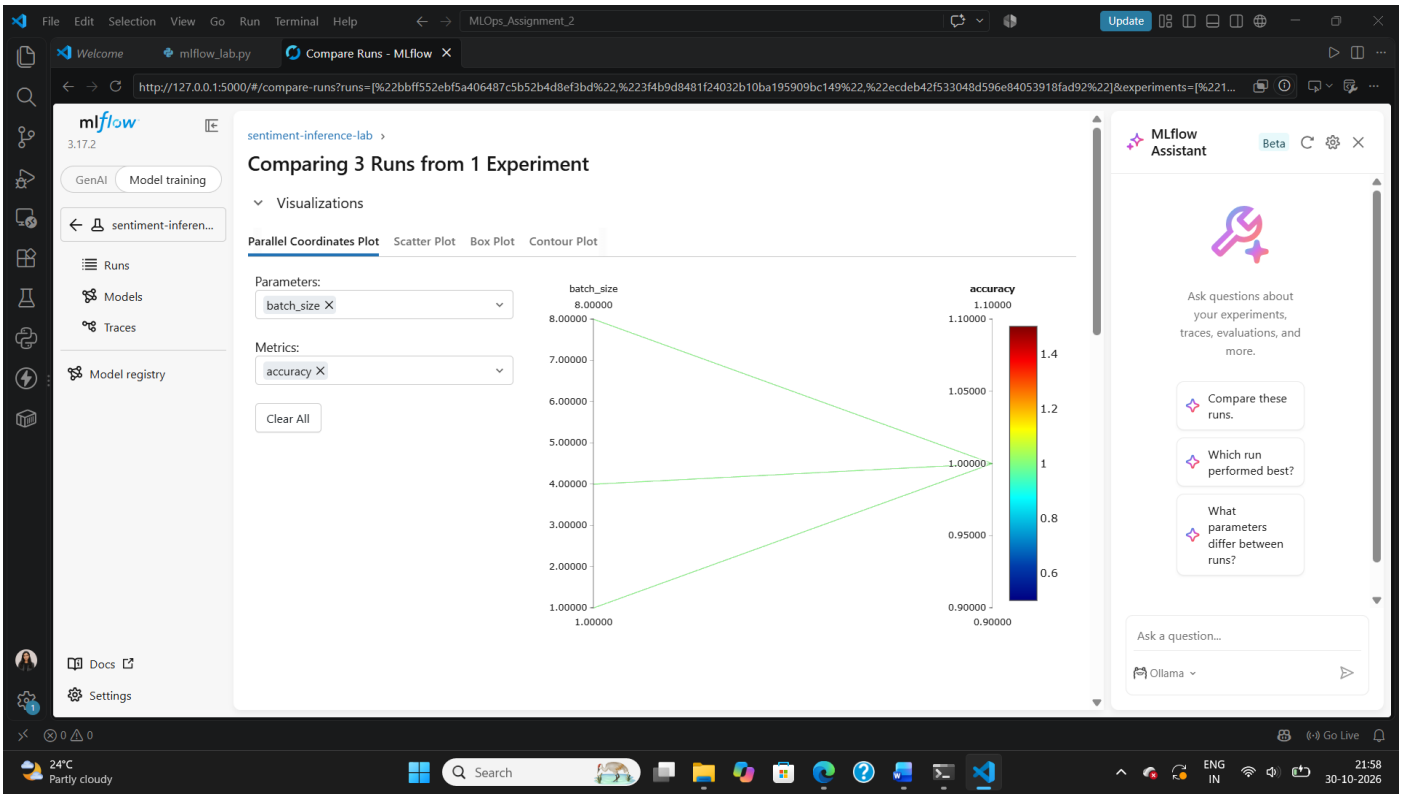
1. Reproducibility and record-keeping: MLflow logs the parameters, metrics, and prediction artifacts for each experiment run, making it easier to review results later.

2. Easy comparison: MLflow provides a dashboard to compare multiple runs side by side using tables and visualizations, making differences in latency, accuracy, and other metrics easier to understand than terminal output.

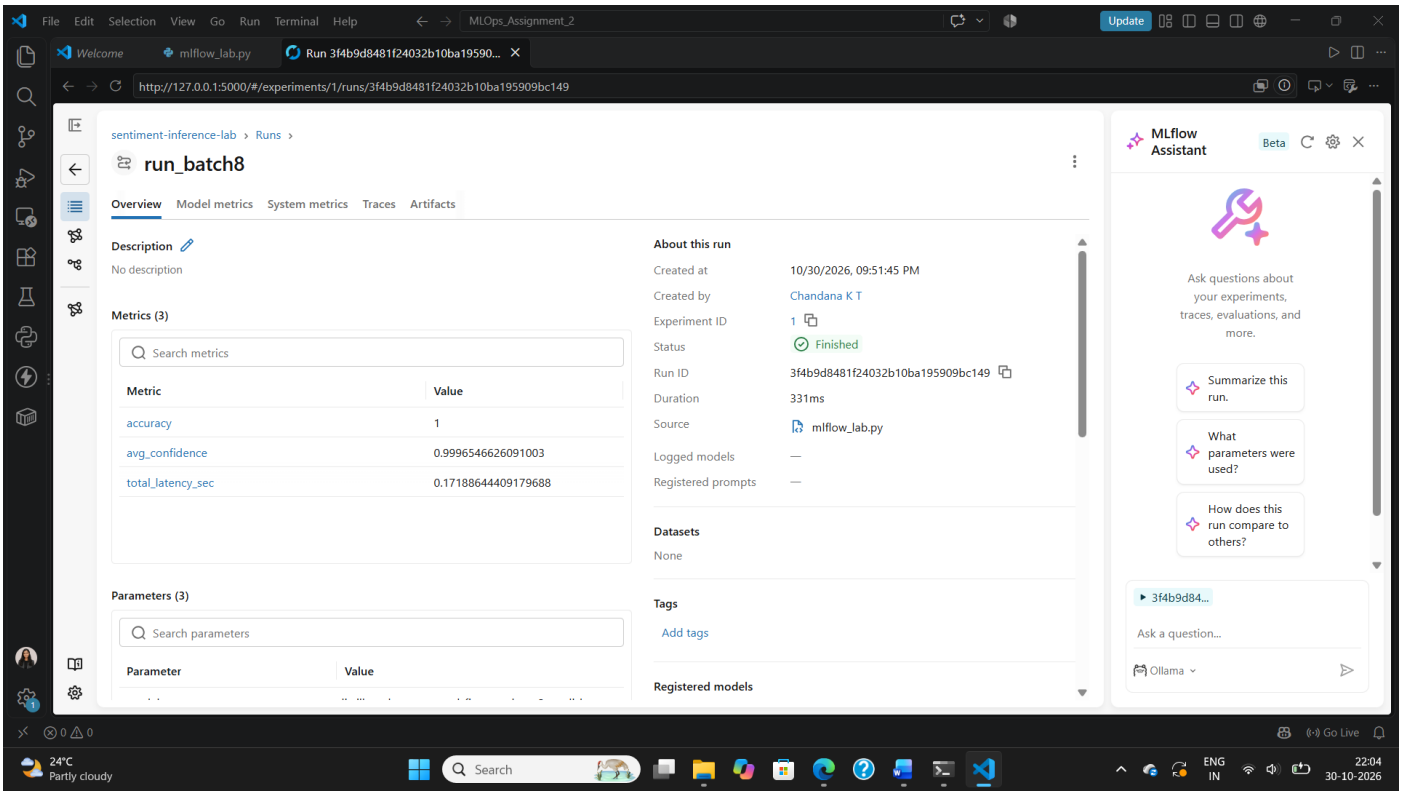


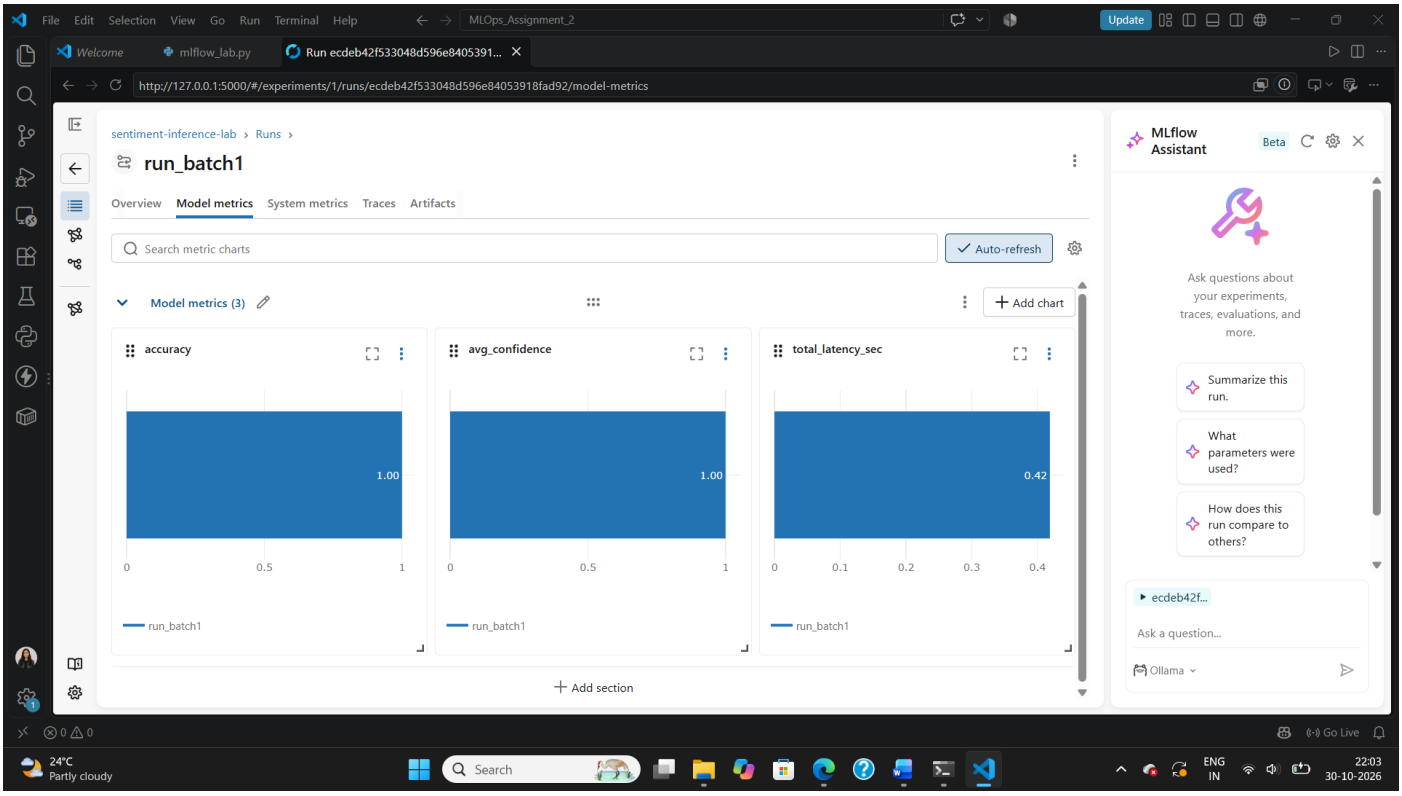
Comparison:

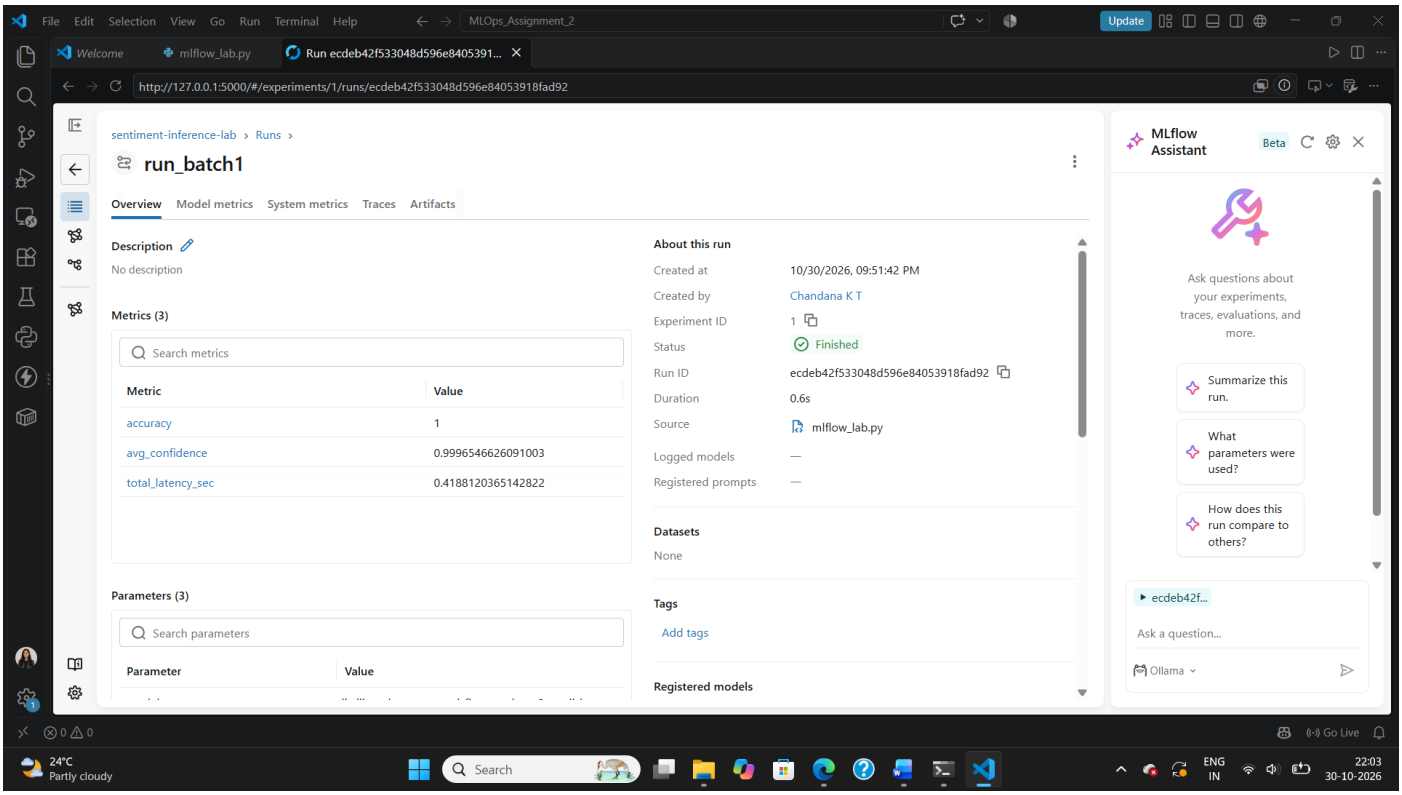

Compare Runs - MLflow
sentiment-inference-lab
Comparing 2 Runs from 1 Experiment
Visualizations
Parallel Coordinates Plot
Scatter Plot
Box Plot
Contour Plot
Parameters:
batch_size
num_samples
model_name
Metrics:
accuracy
avg_confidence
total_latency_sec
Clear All
distilbert
MLflow Assistant
Beta
Ask questions about your experiments, traces, evaluations, and more.
Compare these runs.
Which run performed best?
What parameters differ between runs?
Ask a question...
Ollama

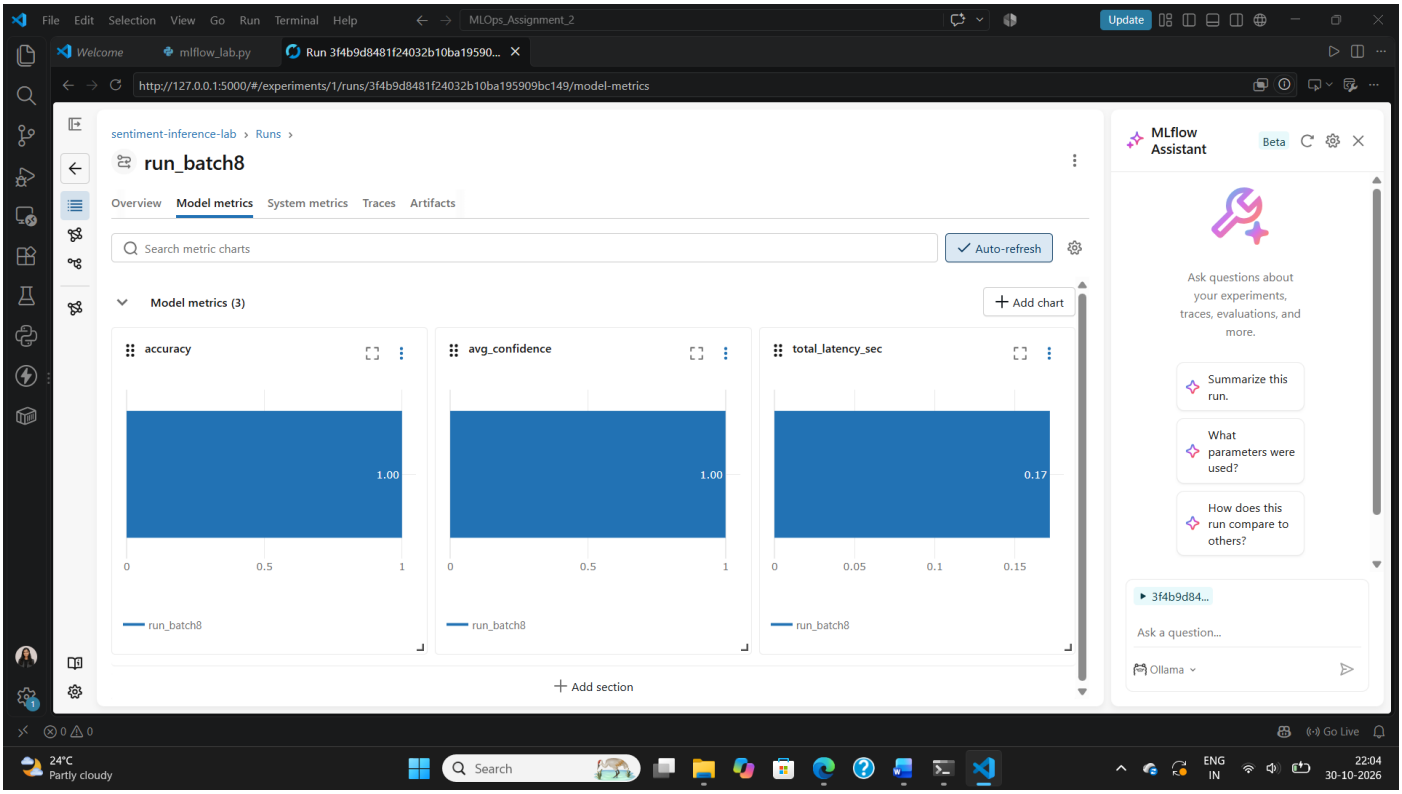

Compare Runs - MLflow
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
sentiment-inference-lab
Comparing 3 Runs from 1 Experiment
Visualizations
Parallel Coordinates Plot
Scatter Plot
Box Plot
Contour Plot
Parameters:
batch_size
Metrics:
accuracy
Clear All
MLflow Assistant
Beta
Ask questions about your experiments, traces, evaluations, and more.
Compare these runs.
Which run performed best?
What parameters differ between runs?
Ask a question...
Ollama

MLflow runs table:

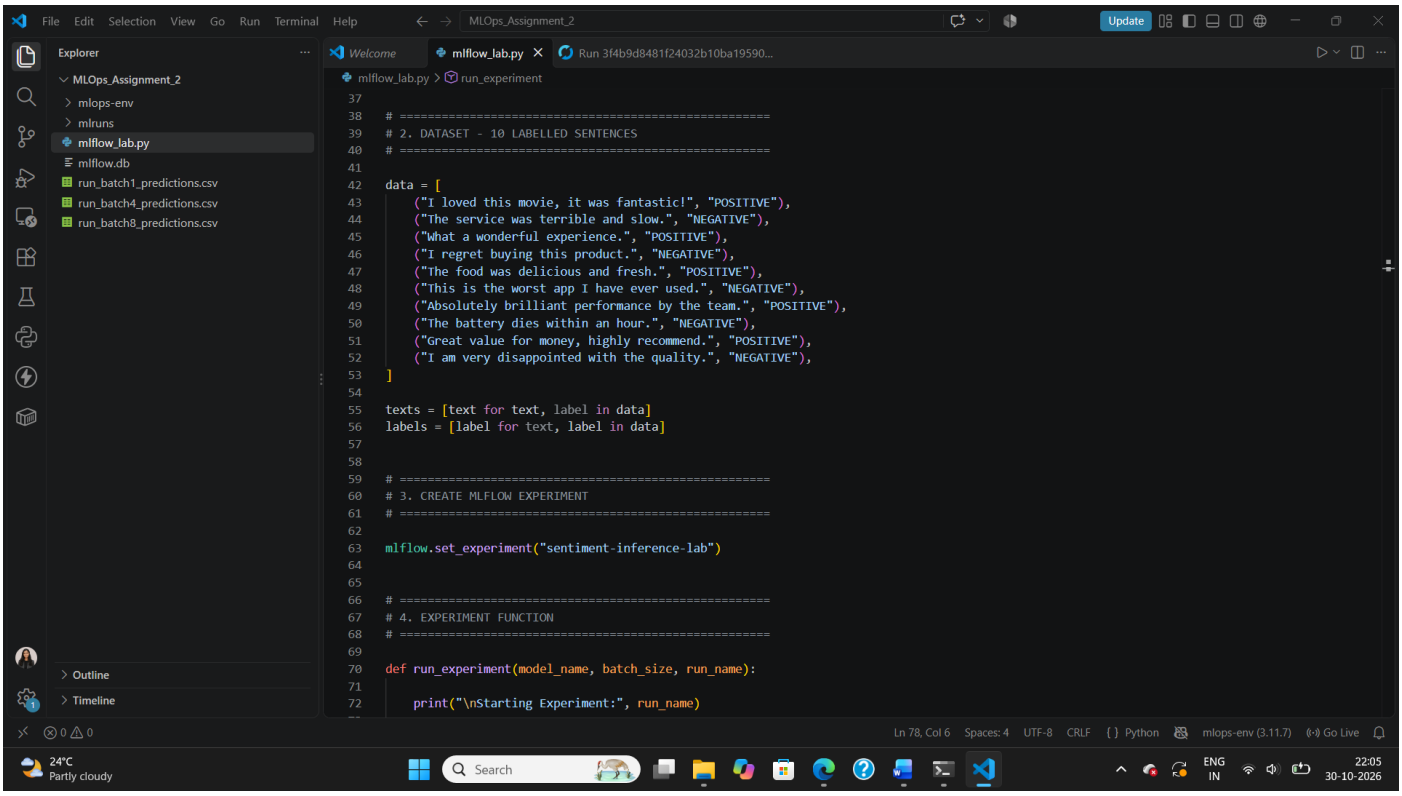





File Edit Selection View Go Run Terminal Help
MLOps_Assignment_2
Update
Welcome
mlflow_lab.py
Run ecdeb42f533048d596e8405391...
http://127.0.0.1:5000/#/experiments/1/runs/ecdeb42f533048d596e84053918fad92
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Description
No description
Metrics (3)
Search metrics
Metric
Value
accuracy
1
avg_confidence
0.9996546626091003
total_latency_sec
0.4188120365142822
Parameters (3)
Search parameters
Parameter
Value
About this run
Created at
10/30/2026, 09:51:42 PM
Created by
Chandana K T
Experiment ID
1
Status
Finished
Run ID
ecdeb42f533048d596e84053918fad92
Duration
0.6s
Source
mlflow_lab.py
Logged models
—
Registered prompts
—
Datasets
None
Tags
Add tags
Registered models
MLflow Assistant
Beta
Ask questions about your experiments, traces, evaluations, and more.
Summarize this run.
What parameters were used?
How does this run compare to others?
ecdeb42f...
Ask a question...
Ollama
24°C
Partly cloudy
Search
ENG
IN
22:03
30-10-2026


File Edit Selection View Go Run Terminal Help
MLOps_Assignment_2
Update
Welcome
mlflow_lab.py
Run 3f4b9d8481f24032b10ba19590...
http://127.0.0.1:5000/#/experiments/1/runs/3f4b9d8481f24032b10ba195909bc149/model-metrics
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.17
0
0.5
1
0
0.05
0.1
0.15
run_batch8
Add section
MLflow Assistant
Beta
Ask questions about your experiments, traces, evaluations, and more.
Summarize this run.
What parameters were used?
How does this run compare to others?
3f4b9d84...
Ask a question...
Ollama
24°C
Partly cloudy
Search
ENG
IN
22:04
30-10-2026

mlflow_lab.py file:



```python

# ====================================================
# 2. DATASET - 10 LABELLED SENTENCES
# ====================================================

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [text for text, label in data]
labels = [label for text, label in data]

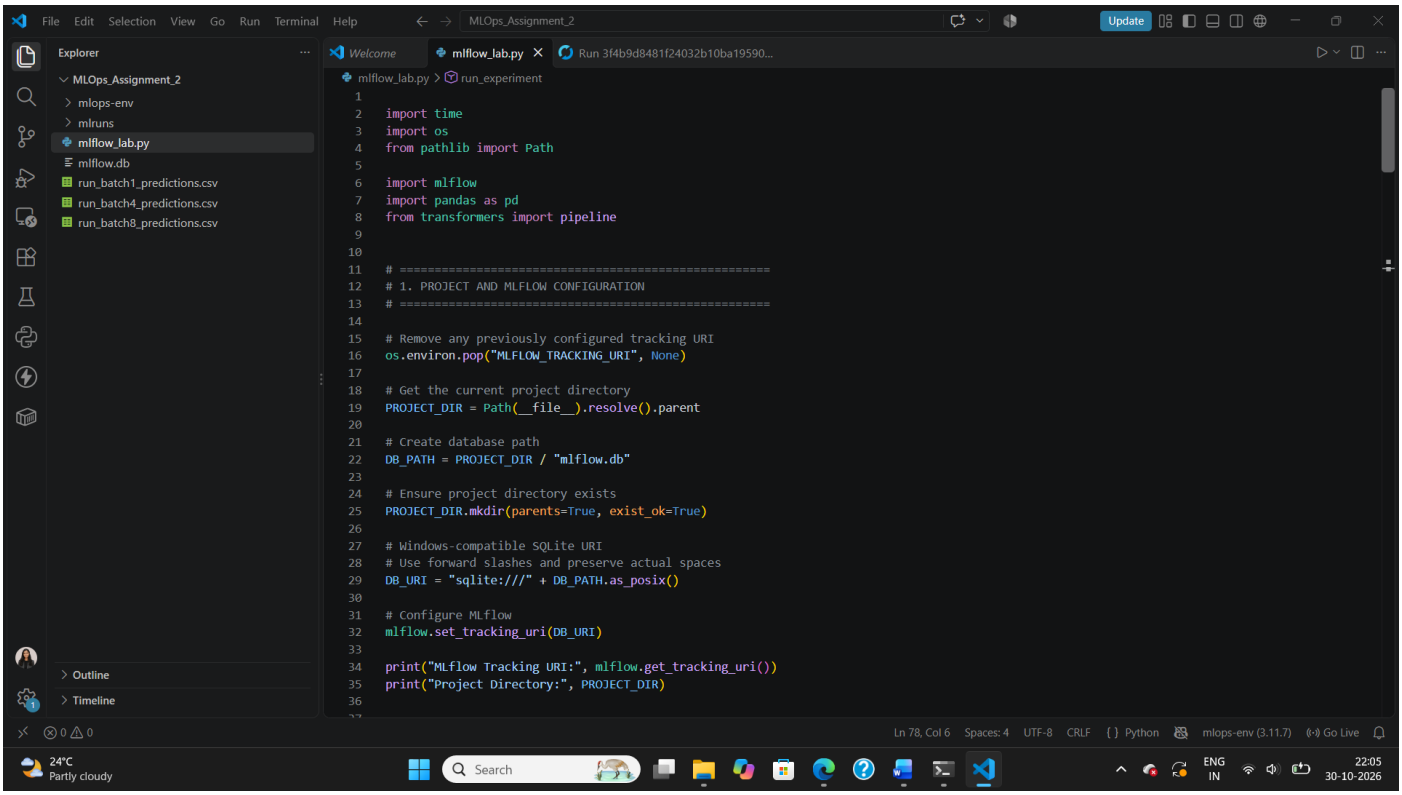

# ====================================================
# 3. CREATE MLFLOW EXPERIMENT
# ====================================================

mlflow.set_experiment("sentiment-inference-lab")


# ====================================================
# 4. EXPERIMENT FUNCTION
# ====================================================

def run_experiment(model_name, batch_size, run_name):

    print("\nStarting Experiment:", run_name)
```



```python

import time
import os
from pathlib import Path

import mlflow
import pandas as pd
from transformers import pipeline


# ====================================================
# 1. PROJECT AND MLFLOW CONFIGURATION
# ====================================================

# Remove any previously configured tracking URI
os.environ.pop("MLFLOW_TRACKING_URI", None)

# Get the current project directory
PROJECT_DIR = Path(__file__).resolve().parent

# Create database path
DB_PATH = PROJECT_DIR / "mlflow.db"

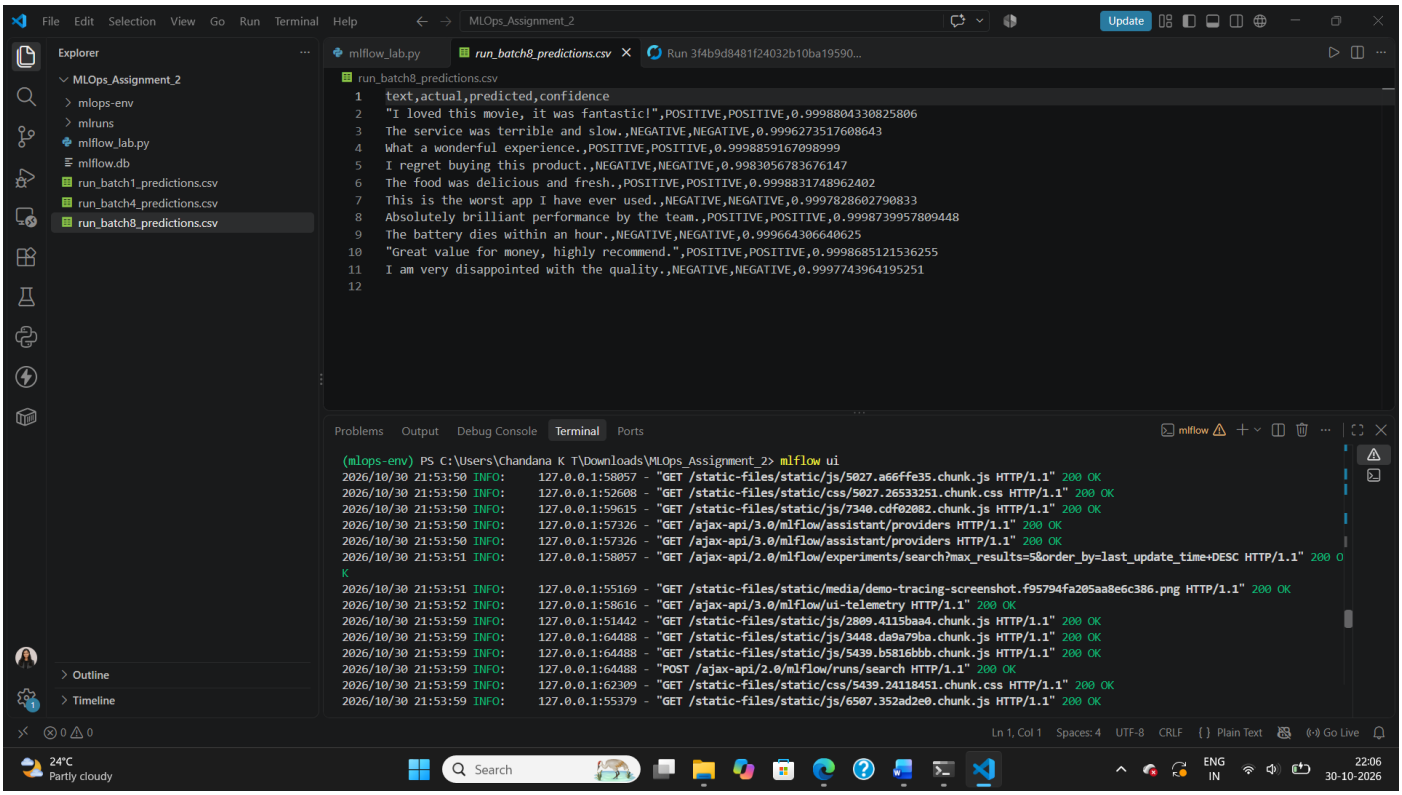
# Ensure project directory exists
PROJECT_DIR.mkdir(parents=True, exist_ok=True)

# Windows-compatible SQLite URI
# Use forward slashes and preserve actual spaces
DB_URI = "sqlite:///" + DB_PATH.as_posix()

# Configure MLflow
mlflow.set_tracking_uri(DB_URI)

print("MLflow Tracking URI:", mlflow.get_tracking_uri())
print("Project Directory:", PROJECT_DIR)
```

Execution :


File Edit Selection View Go Run Terminal Help
MLOps_Assignment_2
Update
Explorer
MLOps_Assignment_2
mlops-env
mlruns
mlflow_lab.py
mlflow.db
run_batch1_predictions.csv
run_batch4_predictions.csv
run_batch8_predictions.csv
mlflow_lab.py
run_batch8_predictions.csv
Run 3f4b9d8481f24032b10ba19590...
run_batch8_predictions.csv
1 text,actual,predicted,confidence
2 "I loved this movie, it was fantastic!",POSITIVE,POSITIVE,0.9998804330825806
3 The service was terrible and slow.,NEGATIVE,NEGATIVE,0.9996273517608643
4 What a wonderful experience.,POSITIVE,POSITIVE,0.9998859167098999
5 I regret buying this product.,NEGATIVE,NEGATIVE,0.9983056783676147
6 The food was delicious and fresh.,POSITIVE,POSITIVE,0.9998831748962402
7 This is the worst app I have ever used.,NEGATIVE,NEGATIVE,0.9997828602790833
8 Absolutely brilliant performance by the team.,POSITIVE,POSITIVE,0.9998739957809448
9 The battery dies within an hour.,NEGATIVE,NEGATIVE,0.999664306640625
10 "Great value for money, highly recommend.",POSITIVE,POSITIVE,0.9998685121536255
11 I am very disappointed with the quality.,NEGATIVE,NEGATIVE,0.9997743964195251
12
Problems Output Debug Console Terminal Ports
mlflow
(mlops-env) PS C:\Users\Chandana K T\Downloads\MLOps_Assignment_2> mlflow ui
2026/10/30 21:53:50 INFO: 127.0.0.1:58057 - "GET /static-files/static/js/5027.a66ffe35.chunk.js HTTP/1.1" 200 OK
2026/10/30 21:53:50 INFO: 127.0.0.1:52608 - "GET /static-files/static/css/5027.26533251.chunk.css HTTP/1.1" 200 OK
2026/10/30 21:53:50 INFO: 127.0.0.1:59615 - "GET /static-files/static/js/7340.cdf02082.chunk.js HTTP/1.1" 200 OK
2026/10/30 21:53:50 INFO: 127.0.0.1:57326 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/30 21:53:50 INFO: 127.0.0.1:57326 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/30 21:53:51 INFO: 127.0.0.1:58057 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+DESC HTTP/1.1" 200 O
K
2026/10/30 21:53:51 INFO: 127.0.0.1:55169 - "GET /static-files/static/media/demo-tracing-screenshot.f95794fa205aa8e6c386.png HTTP/1.1" 200 OK
2026/10/30 21:53:52 INFO: 127.0.0.1:58616 - "GET /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:51442 - "GET /static-files/static/js/2809.4115baa4.chunk.js HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:64488 - "GET /static-files/static/js/3448.da9a79ba.chunk.js HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:64488 - "GET /static-files/static/js/5439.b5816bbb.chunk.js HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:64488 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:62309 - "GET /static-files/static/css/5439.24118451.chunk.css HTTP/1.1" 200 OK
2026/10/30 21:53:59 INFO: 127.0.0.1:55379 - "GET /static-files/static/js/6507.352ad2e0.chunk.js HTTP/1.1" 200 OK
Outline
Timeline
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text Go Live
24°C
Partly cloudy
Search
ENG
IN
22:06
30-10-2026